

JECRC UNIVERSITY JAIPUR

Session -2023-24

Subject: Life Skills-II (DMA011C)

Semester-III

(Common to all disciplines at UG level)

Tutorial Sheet – 04 (CO2)

Unit- 2

Topic- Time and Work

CO2: The students will be able to solve the problems related to time & work and coding decoding.

Work with different efficiencies

Question 1:

Two workers, Alice and Bob, are painting a room. Alice can paint the room in 6 hours, while Bob can do it in 8 hours. If they work together, how long will it take them to paint the room?

- A) 2 hours
- B) 3 hours
- C) 4 hours
- D) 5 hours

Question 2:

A car can travel 60 miles per hour, and a bicycle can travel 15 miles per hour. If they both start at the same time and travel towards the same destination, how much faster is the car?

- A) 15 mph
- B) 30 mph
- C) 45 mph
- D) 50 mph

Question 3:

A gardener can mow a lawn in 2 hours, while their assistant can do it in 3 hours. If they work together, how much of the lawn can they mow in one hour?

- A) $\frac{1}{2}$ of the lawn
- B) $\frac{2}{3}$ of the lawn
- C) $\frac{3}{4}$ of the lawn
- D) None of these

Question 4:

A factory has three machines. Machine X can produce 200 units in an hour, Machine Y can produce 150 units in an hour, and Machine Z can produce 100 units in an hour. If all three machines work simultaneously, how many units can they produce in an hour?

- A) 250 units
- B) 300 units
- C) 400 units
- D) 450 units

Question 5:

A group of students can clean a classroom in 2 hours. If one student leaves, it takes them 3 hours to clean the same classroom. How many students were in the initial group?

- A) 2 students
- B) 3 students
- C) 4 students
- D) 5 students

Question 6:

A chef can bake a cake in 45 minutes, while a novice baker takes 90 minutes to bake the same cake. If they work together, how long will it take them to bake two cakes?

- A) 30 minutes

- B) 45 minutes
- C) 60 minutes
- D) 75 minutes

Question 7:

A printer can print 100 pages in 10 minutes, while another printer can print 150 pages in 15 minutes. If both printers are used simultaneously, how many pages can they print in 20 minutes?

- A) 250 pages
- B) 300 pages
- C) 350 pages
- D) 400 pages

Question 8:

A plumber can fix a leak in 2 hours, while their apprentice can fix it in 3 hours. If they work together, how long will it take them to fix the leak and then another similar leak immediately afterward?

- A) 1 hour
- B) 1.5 hours
- C) 3 hours
- D) 2.5 hours

Pipes and cisterns

Question 9:

A pipe can fill a cistern in 6 hours, while another pipe can empty the same cistern in 8 hours. If both pipes are opened simultaneously, how long will it take to fill the cistern?

- A) 4 hours
- B) 6 hours
- C) 8 hours
- D) 24 hours

Question 10:

A pipe can fill a tank in 4 hours, while another pipe can fill it in 6 hours. If both pipes are opened simultaneously, how long will it take to fill the tank?

- A) 2 hours 24 minutes
- B) 3 hours 24 minutes
- C) 4 hours 24 minutes
- D) 5 hours 24 minutes

Question 11:

A pipe can fill a pool in 10 hours, while another pipe can empty it in 15 hours. If both pipes are opened simultaneously, how long will it take to fill the pool?

- A) 6 hours
- B) 7.5 hours
- C) 9 hours
- D) 15 hours

Question 12:

A pipe can fill a tank in 8 hours, while another pipe can empty it in 12 hours. If both pipes are opened simultaneously, how long will it take to fill half of the tank?

- A) 2 hours
- B) 3 hours
- C) 4 hours
- D) 6 hours

Question 13:

A pipe can fill a reservoir in 5 hours, while another pipe can empty it in 10 hours. If both pipes are opened simultaneously, how long will it take to empty the reservoir?

- A) 2 hours
- B) 4 hours
- C) 5 hours

D) 6 hours

Question 14:

A pipe can fill a tank in 7 hours, while another pipe can empty it in 14 hours. If both pipes are opened simultaneously, how long will it take to fill the tank to $\frac{2}{3}$ of its capacity?

A) 3 hours

B) 4 hours

C) 5 hours

D) 6 hours

Question 15:

A pipe can fill a pool in 9 hours, while another pipe can empty it in 6 hours. If both pipes are opened simultaneously, how long will it take to fill the pool if it initially contains $\frac{3}{4}$ of its capacity?

A) 2 hours

B) 3 hours

C) 4 hours

D) 5 hours

Question 16:

A pipe can fill a tank in 15 hours, while another pipe can empty it in 20 hours. If both pipes are opened simultaneously, how long will it take to fill the tank to $\frac{3}{5}$ of its capacity?

A) 5 hours

B) 6 hours

C) 7 hours

D) 8 hours

Work equivalence

Question 17:

If you push a 50 kg box horizontally with a force of 200 N for a distance of 5 meters, how much work have you done?

- A) 100 J
- B) 500 J
- C) 1,000 J
- D) 2,000 J

Question 18:

A person lifts a 20 kg bag of rice vertically to a height of 2 meters. How much work is done against gravity?

- A) 20 J
- B) 40 J
- C) 80 J
- D) 200 J

Question 19:

A car accelerates from rest to a speed of 30 m/s in 5 seconds. What is the work done by the car's engine during this acceleration?

- A) 750 J
- B) 1,500 J
- C) 3,000 J
- D) none of these

Question 20:

A person applies a force of 50 N to lift a box with a weight of 500 N vertically upwards. How much work is done by the person if the box is lifted 2 meters?

- A) 50 J
- B) 100 J

C) 200 J

D) 500 J

Question 21:

A machine does 2,000 J of work in lifting a crate to a height of 4 meters. What force did the machine apply to lift the crate?

A) 200 N

B) 400 N

C) 500 N

D) 800 N

Question 22:

A motor does 600 J of work in lifting a load. If it takes 10 seconds to perform this work, what is the power of the motor?

A) 6 W

B) 60 W

C) 120 W

D) 300 W

Question 23:

A cyclist generates 200 W of power for 2 hours. How much work did the cyclist do during this time?

A) 200 J

B) 400 J

C) 1,200 J

D) 2,400 J

Question 24:

A machine applies a force of 300 N to move an object horizontally for a distance of 10 meters. If the machine exerts this force for 5 seconds, what is the power output of the machine?

- A) 15 W
- B) 30 W
- C) 150 W
- D) 600 W

Division of wages

Question 25:

Three workers, A, B, and C, earn a total of \$1,200 per week. If A earns \$400, and B earns \$300, how much does C earn?

- A) \$200
- B) \$300
- C) \$500
- D) \$600

Question 26:

A company pays its employees \$15 per hour. If an employee works for 40 hours in a week, how much will they earn in that week?

- A) \$400
- B) \$500
- C) \$600
- D) \$700

Question 27:

Two friends, X and Y, work at the same company. X earns \$500 per week, while Y earns \$15 per hour and works for 30 hours a week. Who earns more in a week?

- A) X
- B) Y
- C) They earn the same amount
- D) Need more information to determine

Question 28:

A team of four workers completes a project in 12 days. If they are paid a total of \$6,000 for the project, how much does each worker earn per day?

- A) \$20
- B) \$25
- C) \$30
- D) \$40

Question 29:

In a restaurant, the total tips collected in a day amount to \$500. If there are 5 waiters, how should the tips be divided equally among them?

- A) \$100 each
- B) \$200 each
- C) \$800 each
- D) \$110 each

Question 30:

A construction project requires three different types of workers: carpenters, plumbers, and electricians. If the total budget for the project is \$30,000, and it is decided to allocate 40% to carpenters, 30% to plumbers, and the rest to electricians, how much will be allocated to electricians?

- A) \$6,000
- B) \$7,000
- C) \$8,000
- D) \$9,000

Question 31:

A company divides its profits equally among its three partners. If they share a total profit of \$45,000, how much does each partner receive?

- A) \$10,000
- B) \$15,000

C) \$20,000

D) \$25,000

Question 32:

In a project, two contractors, X and Y, share the earnings in a 2:3 ratio, respectively. If they earn a total of \$10,000, how much does X earn?

A) \$2,000

B) \$3,000

C) \$4,000

D) \$5,000